Completeness of the Milky Way globular-cluster catalogue from the phases of cluster orbits

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Abstract

We estimate how the completeness of the Milky Way globular-cluster (GC) catalogue depends on absolute magnitude and heliocentric distance, using the 165 GCs with full six-dimensional phase-space data. The estimator conditions on each cluster's own orbit: if clusters are phase-mixed, the present position of a detected cluster samples its orbit with probability proportional to the completeness C at that point. Maximising the resulting likelihood needs no model of the intrinsic GC density. On mock catalogues with known selection it recovers completeness without bias and with calibrated intervals (recovered/true = 0.97–0.99, 68% coverage 0.67–0.71). A fixed-target orbit-superposition estimator in the spirit of Khoperskov et al. (2025) is biased high by $\approx 25\%$ on the same mocks. Applied to the data with C a function of extinction-free apparent magnitude V, completeness falls from 1 at $V < 6$ (assumed) to 0.14 at $V > 11$. Weighting each cluster by the inverse of its orbit-averaged completeness turns 165 observed GCs into 483 (68%: 335–765); the faint clusters ($M_V > -6.5$) rise from 55 to 267 (168–473). Mocks show these totals are lower limits, because faint clusters on orbits that never enter the detectable volume cannot be recovered by reweighting observed orbits. A first extinction test raises the total by $\approx 40\%$ but is disfavoured by the data, so extinction remains the leading open systematic.

1 Data

We use the 165 confirmed GCs with full phase-space data (positions, distances, proper motions, line-of-sight velocities and their errors) from the PI's compilation of Vasiliev & Baumgardt (2021) astrometry. Absolute magnitudes come from the Baumgardt combined table[1] as $M_V = 4.83 - 2.5\log_{10}(M_{\rm dyn}/\Upsilon_V)$, with $M_{\rm dyn}$ the dynamical mass and Υ_V the V-band mass-to-light ratio. Two confirmed clusters without 6D data are excluded. Progenitor labels (in-situ: main bulge and main disc, 61 GCs; accreted: Gaia-Enceladus, Sequoia, Helmi streams, high-energy group and Sagittarius, 64 GCs) come from the catalogue column `PROG_MA`; we have not verified their provenance (they appear to follow Massari et al. 2019).

Positions and velocities are transformed to the Galactocentric frame with the astropy v4.0 solar parameters ($R_0 = 8.122$ kpc, $z_\odot = 20.8$ pc, $\boldsymbol{v}_\odot = (12.9, 245.6, 7.78)$ km s^{-1}; Astropy Collaboration 2022). Orbits are integrated with AGAMA (Vasiliev 2019) in a static axisymmetric potential (Miyamoto–Nagai disc, $6.8\times10^{10}\,M_\odot$, $a = 3.0$, $b = 0.28$ kpc; two Dehnen spheroids, $5\times10^9\,M_\odot$ at 1.0 kpc and $1.71\times10^9\,M_\odot$ at 0.07 kpc; NFW halo, $5.4\times10^{11}\,M_\odot$, $r_s = 15.62$ kpc), forward for 5 Gyr from the present and sampled at 500 equal time steps. Energy is conserved to 2×10^{-6}. No observed cluster and no orbit sample lies beyond 150 kpc.

2 Method

2.1 Per-cluster phase likelihood

Let cluster i move on a known orbit, sampled at times t, with heliocentric distance $d_i(t)$. If the population is phase-mixed in a static potential, the present phase of an undetected cluster is uniform

[1] Downloaded 2026-10-09 from https://people.smp.uq.edu.au/HolgerBaumgardt/globular/combined_table.txt.

in t. Detection with probability C tilts that distribution, so the present phase of a *detected* cluster, conditional on its orbit, has density proportional to C at that phase. The log-likelihood of the observed present positions is

$$\ln \mathcal{L} = \sum_i \Big[\ln C(M_{V,i}, d_{i,\mathrm{now}}) - \ln \langle C(M_{V,i}, d_i(t)) \rangle_t \Big]. \tag{1}$$

Two properties make Eq. (1) useful. First, no model of the intrinsic GC density enters. Second, conditioning on each detected orbit removes the dependence on which orbits happen to be in the catalogue. The overall scale of C cancels, so the data constrain C only up to one constant, and only over the range of d (or V) that the observed orbits span.

We take C piecewise constant in bins, write $\theta_k = \ln(C_k/C_0)$ with the reference bin $k = 0$ fixed, and maximise Eq. (1) with L-BFGS-B ($-8 \le \theta_k \le 4$). The likelihood is concave in θ. Intervals are 68% profile-likelihood intervals ($2\Delta \ln \mathcal{L} = 1$). The last bin is open-ended.

2.2 Two parameterisations of C

$C(M_V, d)$ per luminosity class. We split the sample into three equal-count M_V classes ($\approx$ 55 GCs each) and, within each, into equal-count distance bins of at least 15 GCs (three bins of 18–19 GCs). Bins are chosen once on the observed sample and then frozen. Each class yields $C(d)/C(\text{nearest bin})$; the classes are not linked to one another.

$C(V)$. To link the classes we assume completeness depends on apparent magnitude only, $C = C(V)$ with $V = M_V + 5\log_{10}(d/10\,\mathrm{pc})$. This is a special case of $C(M_V, d)$: a faint cluster nearby and a bright cluster farther away with the same V share the same completeness. All 165 clusters then enter one fit, with $V_i(t)$ varying along each orbit. We use eight equal-count bins in observed V (20–21 GCs each) and fix the brightest bin ($V < 6.0$) as complete, $C_0 = 1$. This single anchor makes C absolute at all M_V. V excludes extinction; Sec. 4.5 tests one extinction model.

2.3 Corrected counts

Each observed cluster receives the weight $w_i = 1/q_i$, with $q_i = \langle C(V_i(t)) \rangle_t$ its orbit-averaged completeness. Corrected distributions are weighted histograms. This is the Horvitz–Thompson (1952) inverse-inclusion-probability estimator with the inclusion probability averaged over orbital phase. It corrects for incompleteness along the orbits of observed clusters; it cannot add clusters on orbit families that are never detectable. Uncertainties on corrected counts come from a cluster bootstrap (200 replicates): each replicate resamples clusters, refits C on the fixed bins, and sums the refitted weights of the resampled clusters.

2.4 Symmetry tests

If the model holds, the present position of each cluster is a draw from its orbit's C-weighted phase distribution. For three statistics — z, y (the Galactic-longitude side) and the azimuth from the Sun's direction $|\phi'|$ — we compute each cluster's randomised probability integral transform (PIT) against that distribution, pool the 165 values, and form a Kolmogorov–Smirnov statistic. The p-value comes from 500 null simulations that redraw present phases from the fitted C and refit C in every simulation, with a +1 correction.

2.5 Comparison estimator: orbit superposition with a fixed target

Following Khoperskov et al. (2025), we also fit non-negative orbit weights so that the time-averaged occupancy of the observed orbits on an $(R, |z|)$ grid reproduces a fixed target density (spherical broken power law with inner slope 2, outer slope 4, $r_b = 5$ kpc, $r_c = 0.3$ kpc; and variants), and take $C = N_{\mathrm{obs}}/N_{\mathrm{corr}}$ in each distance bin. Unlike Eq. (1), this estimator depends on the assumed target density.

3 Validation on mock catalogues

Mock populations are drawn from an isotropic quasi-spherical distribution function in the same potential, with a broken-power-law density (inner slope 2, outer slope 4, $r_b = 5$ kpc, $r_c = 0.3$ kpc), 250 GCs per realisation and a Gaussian luminosity function $M_V \sim \mathcal{N}(-7.4, 1.2)$. Each mock cluster sits at a uniformly random phase of its orbit and is detected with probability $C_{\rm true} = \{1+\exp[(M_V-(V_{\rm lim}-{\rm DM}))/0.5]\}^{-1}$, $V_{\rm lim} = 8$, so $C_{\rm true}$ is a function of V. Scenarios B–D break the assumptions: B multiplies C by a latitude term $1 - 0.7\,e^{-|b|/5^\circ}$; C adds a non-phase-mixed clump of 15 Sagittarius-like clusters; D makes C depend on M_V and d but not through V alone. Each estimator is compared with its own correct truth: for the phase likelihood, the value it converges to for an infinite sample; for the target estimator, the true detected-to-intrinsic ratio per bin.

Table 1: Mock recovery of relative completeness $C(d)/C(\text{nearest bin})$ per luminosity class (100 realisations per scenario). Ratios are recovered/true, median and 16–84% range; coverage is the fraction of nominal 68% intervals containing the truth.

Estimator	A: assumptions hold	B: latitude-dependent C	C: Sgr-like clump
Phase likelihood	0.99 (0.68–1.41); cov. 0.67	0.98 (0.71–1.52); 0.71	0.97 (0.68–1.34); 0.71
Target, plain ratio	1.29 (0.88–1.85); 0.48	1.30; 0.54	1.39; 0.45
Target, bias-reduced	1.23 (0.84–1.77); 0.50	1.22; 0.58	1.32; 0.47

Table 2: Mock validation of $C(V)$ and the corrected counts (100 realisations per scenario).

	A: $C(V)$ true	B: latitude	C: Sgr clump	D: not $C(V)$
$C(V)$ shape, recovered/true	0.92	1.08	1.01	1.08
68% coverage	0.69	0.68	0.68	0.67
Corrected total / true	0.80 (0.60–1.07)	0.56	0.71	0.87
Faint ($M_V > -6.5$): observed / true	0.15	0.12	0.15	0.64
Faint: corrected / true	0.44	0.27	0.42	0.73

Tables 1 and 2 and Figs. 1 and 2 summarise the tests. The phase likelihood recovers completeness without bias and with calibrated intervals in every scenario, including those that break its assumptions. The target estimator is biased high by about 25% even when given the correct target. The corrected totals are low: in scenario A, where the model is correct, they recover 80% of the true population and 44% of the faint clusters. The shortfall comes from faint clusters whose orbits never bring them within detectable distance; no reweighting of the observed orbits recovers them. The bootstrap intervals on the total match the scatter across realisations (half-width $\approx 26\%$ of truth, scatter 24%), so their low coverage reflects this bias, not mis-sized intervals.

Two diagnostics have little power at this sample size. The symmetry tests reject at $p < 0.05$ in 6–13% of null mocks (the $|\phi'|$ test reaches $p \leq 0.01$ in 2%) and in 2–19% of mocks that break the assumptions. The AIC comparison of $C(V)$ against free per-class $C(d)$ favours the free model in only 31% of scenario-D mocks.

4 Results

4.1 Completeness per luminosity class

Table 3 and Fig. 3 give the per-class completeness. In the two brighter classes completeness stays near unity out to ≈ 10 kpc and falls to 0.4–0.6 beyond. The faint class drops to 0.44 already at 8–21 kpc. In-situ clusters lose completeness steeply with distance (0.18 beyond 8.8 kpc); accreted clusters stay near unity to 27 kpc and fall to 0.27 beyond (Fig. 4). The fixed-target estimator agrees with the phase likelihood within the intervals in most bins, but for the accreted clusters it gives

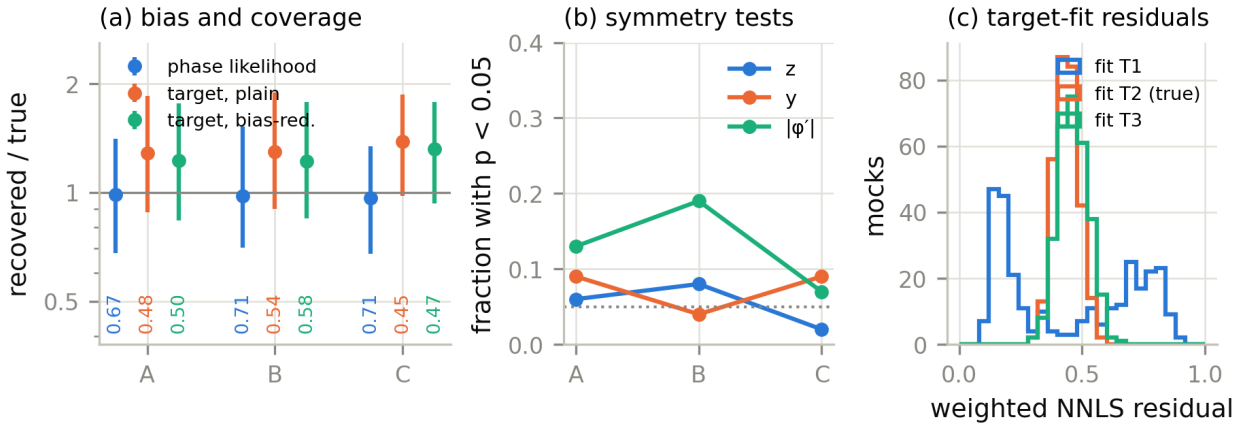


Figure 1: Mock validation of the per-class estimators. (a) Recovered/true relative completeness (median and 16–84% range) for the phase likelihood and the target estimator, plain and bias-reduced, in scenarios A–C; the vertical numbers give the 68% coverage. (b) Fraction of mocks in which each symmetry test gives $p < 0.05$; the dotted line marks 5%. (c) Target-fit residuals when the truth is T2; a residual cannot rank targets, since a T1 fit often beats the true T2.

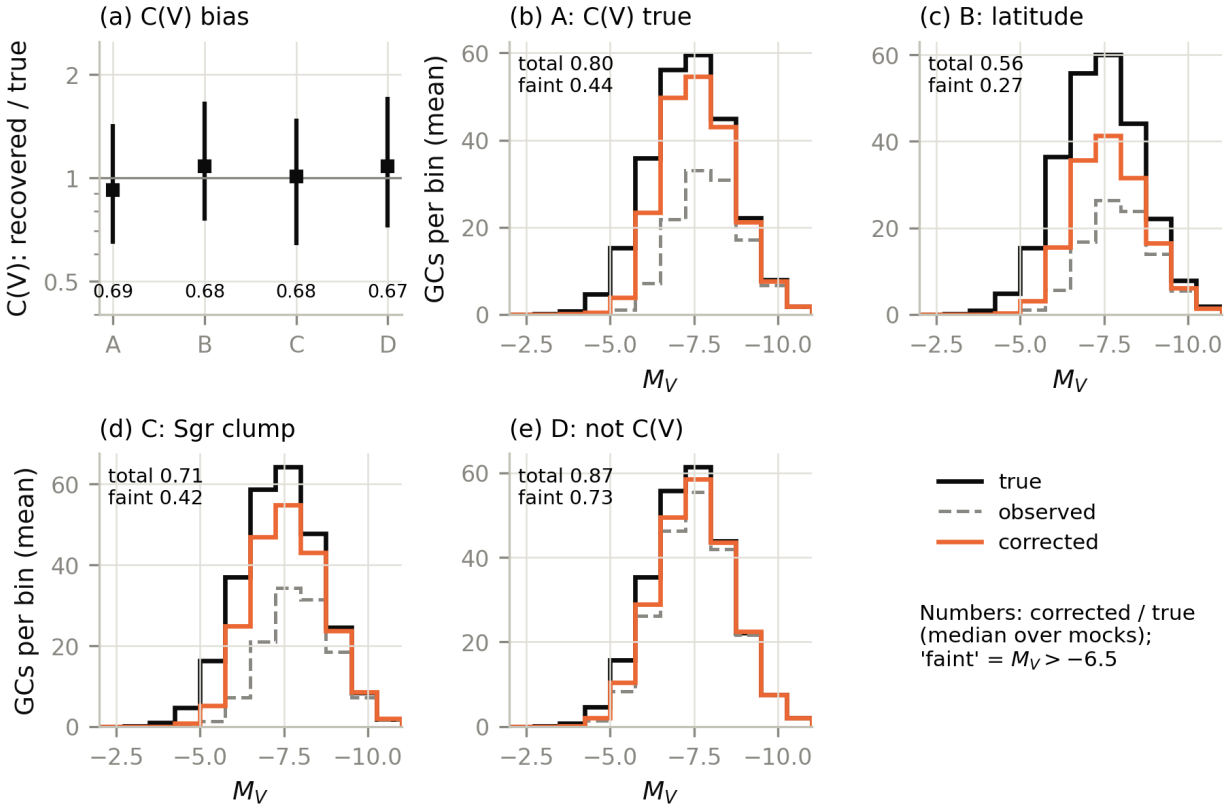


Figure 2: Mock validation of $C(V)$. (a) Recovered/true $C(V)$ per bin (median and 16–84% range) in scenarios A–D; numbers give the 68% coverage. (b)–(e) True (black), observed (dashed) and corrected (orange) M_V distributions, mean over realisations; numbers give the median corrected/true ratio for all GCs and for $M_V > -6.5$.

values above 1 beyond 18 kpc, which the phase likelihood does not reproduce; these follow from the assumed target shape.

Table 3: Phase-likelihood completeness relative to the nearest distance bin, with 68% profile intervals. Each class holds 55 GCs in three bins of 18–19.

M_V class	nearest bin	second bin	third bin
-10.54 to -7.58	0–8.3 kpc: 1	8.3–11.1 kpc: 1.22 [0.88, 1.71]	>11.1 kpc: 0.63 [0.41, 0.95]
-7.58 to -6.50	0–7.4 kpc: 1	7.4–9.5 kpc: 0.75 [0.52, 1.07]	>9.5 kpc: 0.41 [0.28, 0.61]
-6.50 to -0.96	0–7.8 kpc: 1	7.8–21.5 kpc: 0.44 [0.31, 0.62]	>21.5 kpc: 0.56 [0.27, 1.22]

4.2 Symmetry tests

The pooled PIT tests (Fig. 5) give $p = 0.46$ for z, 0.042 for y and 0.006 for $|\phi'|$. Split by population, the in-situ clusters give $p = 0.040$, 0.29 and 0.006, and the accreted clusters 0.51, 0.49 and 0.75. Since null mocks reach $p \leq 0.01$ for $|\phi'|$ in only 2% of realisations, the in-situ clusters carry an azimuthal structure that completeness depending on (M_V, d) alone does not explain. The Galactic bar, which this analysis does not mask, is one candidate. The data do not identify the cause.

4.3 Completeness versus apparent magnitude

Table 4 and Fig. 6 give $C(V)$. Completeness falls steadily from unity at $V < 6$ to 0.14 [0.07, 0.30] at $V > 11$. The per-class completeness curves implied by $C(V)$ follow the free per-class fits (Fig. 6, right), and the AIC prefers $C(V)$ (-6.0 against -1.5), although the mocks show this comparison is weak.

Table 4: $C(V)$ from the phase likelihood (brightest bin fixed at 1); 68% profile intervals. Each bin holds 20–21 GCs.

V	<6.0	6.0–6.6	6.6–7.2	7.2–7.6	7.6–8.2	8.2–9.2	9.2–11.0	>11.0
C	1	1.17	0.56	0.64	0.41	0.25	0.21	0.14
68%	–	[0.82, 1.68]	[0.38, 0.82]	[0.43, 0.95]	[0.27, 0.62]	[0.16, 0.39]	[0.13, 0.35]	[0.07, 0.30]

4.4 Corrected luminosity and metallicity distributions

Weighting by $1/q_i$ (weights from 0.94 to 6.9) turns the 165 observed GCs into 483 (68%: 335–765). The correction grows with faintness (Fig. 7): bright clusters ($M_V < -8$) rise from 40 to 60, intermediate ones ($-8 \leq M_V < -6.5$) from 70 to 156, and faint ones ($M_V \geq -6.5$) from 55 to 267 (168–473). The corrected metallicity distribution is about three times the observed one at all [Fe/H] and keeps its bimodal shape, with peaks near -1.5 and -0.6. Given Sec. 3, these corrected numbers are lower limits if the real selection resembles the mocks.

4.5 Sensitivity to extinction

Adding $A_V = 3.1\,E(B-V)$, with each cluster's present-day reddening held fixed along its orbit, changes the totals substantially: 690 (457–1172) in total and 337 (203–652) faint (Fig. 8). The data disfavour this model, however: its AIC is $+1.3$, against -6.0 without extinction and -1.5 for free per-class fits. Holding present reddening fixed as a cluster moves to other sightlines is unphysical, so this test shows that extinction matters at the 40% level but does not measure its effect.

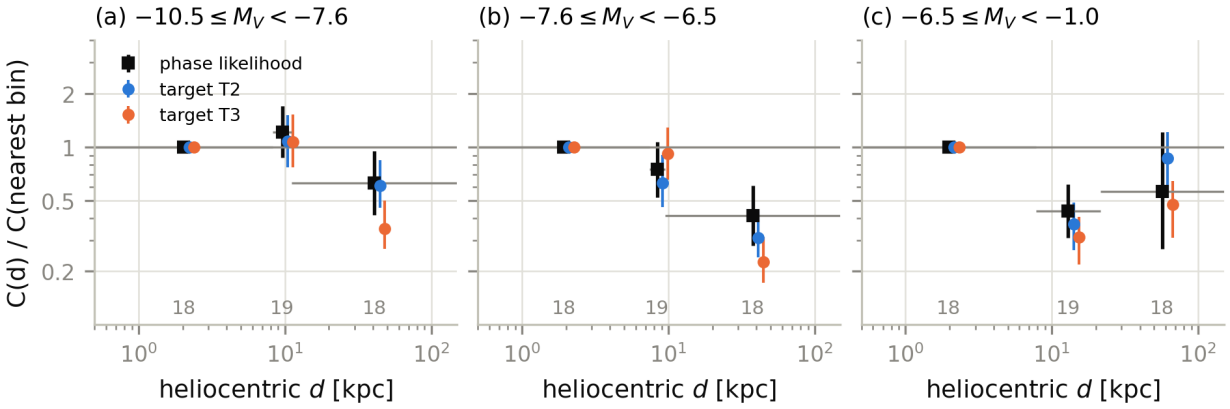


Figure 3: Completeness relative to the nearest distance bin for three M_V classes: phase likelihood (black squares, 68% profile intervals; grey bars show the bin extent) and the fixed-target estimator for targets T2 and T3 (coloured; bias-reduced, 68% bootstrap). Numbers give the observed clusters per bin.

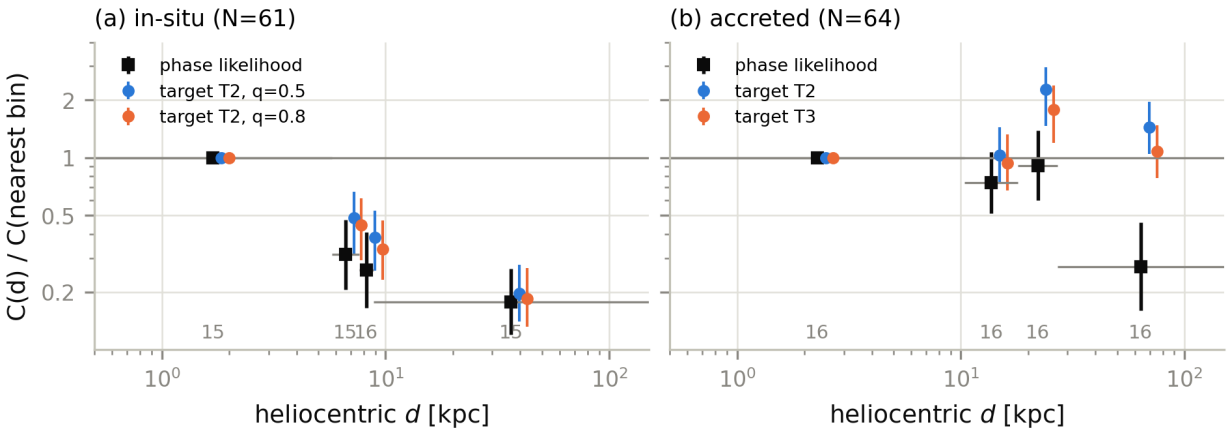


Figure 4: As Fig. 3, for in-situ (left) and accreted (right) clusters, all luminosities combined.

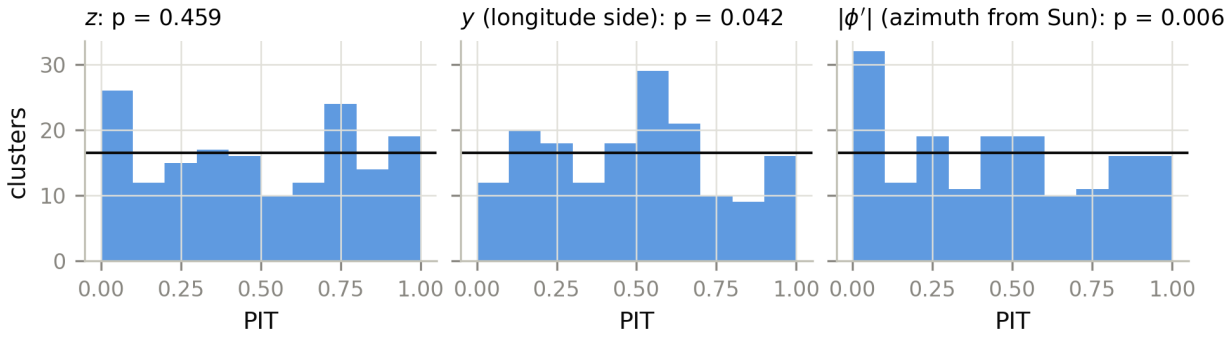


Figure 5: Pooled PIT values for z, y and $|\phi'|$ against each orbit's C-weighted phase distribution. Uniform histograms are expected if the model holds; the titles give the KS statistic and the null-calibrated p-value.

5 Limitations and next steps

1. **Unseen orbit families.** The corrected counts are lower limits (Table 2). Recovering faint clusters on never-detectable orbits requires an extra assumption — that faint and bright GCs share one orbit distribution — after which each faint cluster can be weighted by the fraction of the orbit population from which it would be visible, in the spirit of $1/V_{\max}$ (Schmidt 1968). Such a total must be reported separately and tested against corrected orbit distributions of bright and faint clusters, by population.

2. **Extinction.** Evaluate a 3D dust map along each orbit and add A_V to V; test on mocks with direction-dependent extinction.

3. **Error propagation.** The intervals cover sampling noise only. An ensemble of perturbed phase-space coordinates, alternative potentials and longer, denser integrations would give the systematic spread.

4. **Smooth $C(V)$ and realistic mocks.** A two- or three-parameter monotone $C(V)$ may tighten the results; one bin currently exceeds the anchor (1.17). Mocks matched to the observed sample size, magnitude–distance distribution and extinction would make the validation more direct.

5. **Anchor and assumptions.** All absolute numbers assume the brightest-V bin is complete. The method assumes phase mixing in a static axisymmetric potential; the in-situ azimuthal signal shows where that may fail.

References

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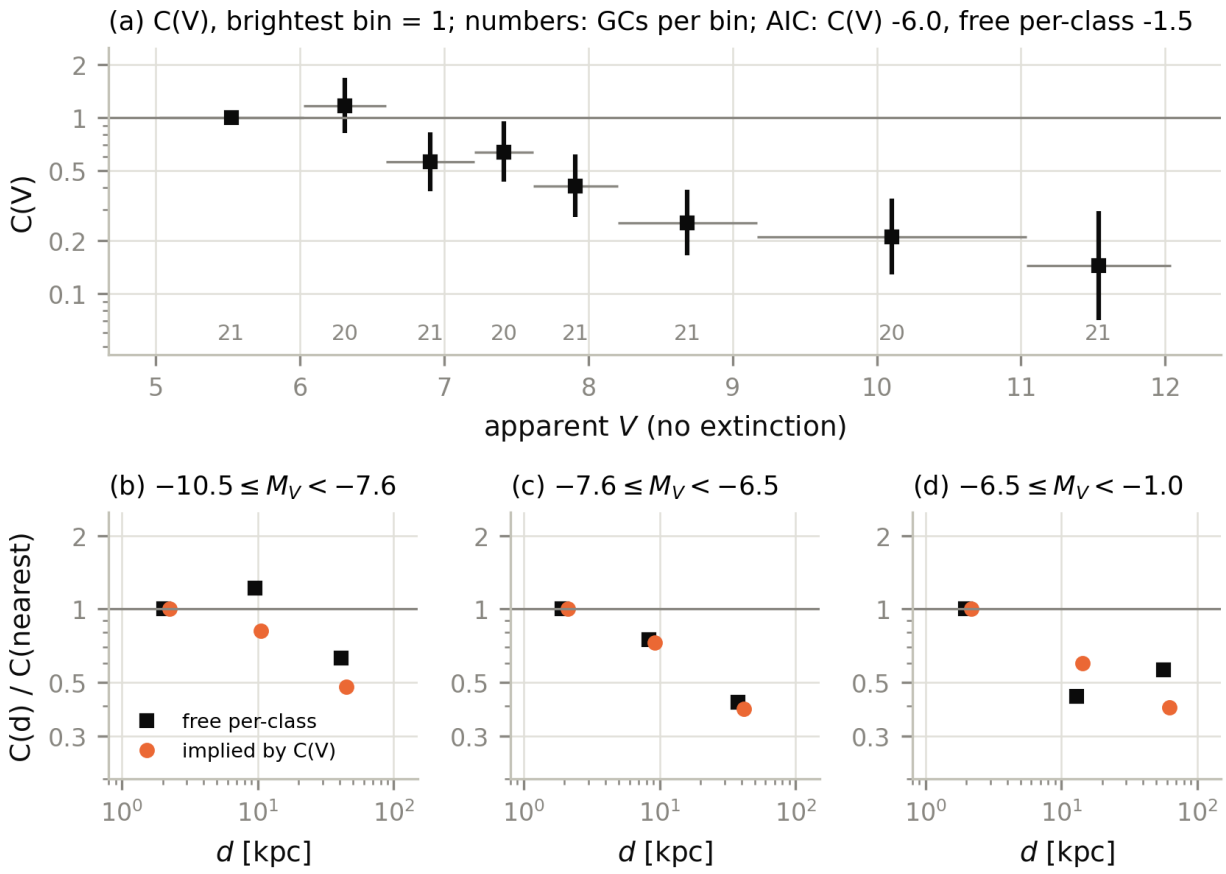


Figure 6: (a) $C(V)$ with 68% profile intervals; grey bars show the bin extent and numbers give clusters per bin. (b)–(d) Per-class completeness versus distance from free fits (black) and implied by $C(V)$ (orange).

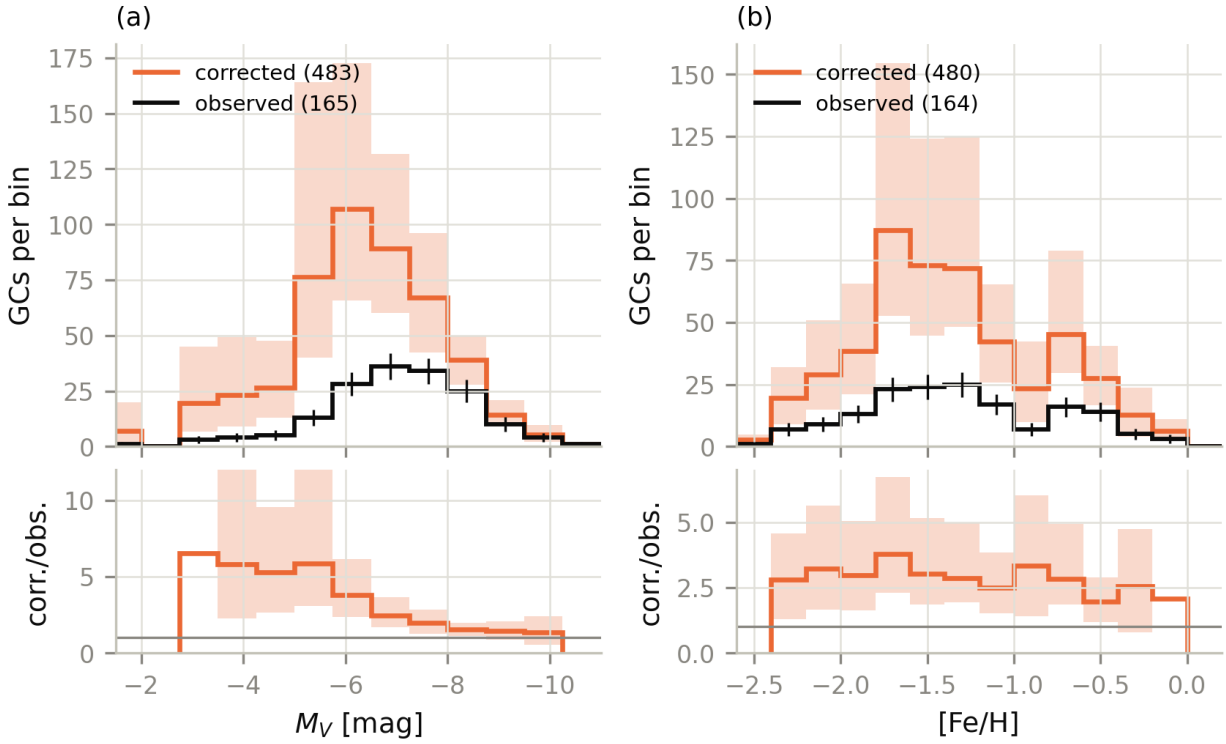


Figure 7: Observed (black) and completeness-corrected (orange, 68% bootstrap band) M_V and [Fe/H] distributions using $C(V)$. Lower panels: corrected/observed.

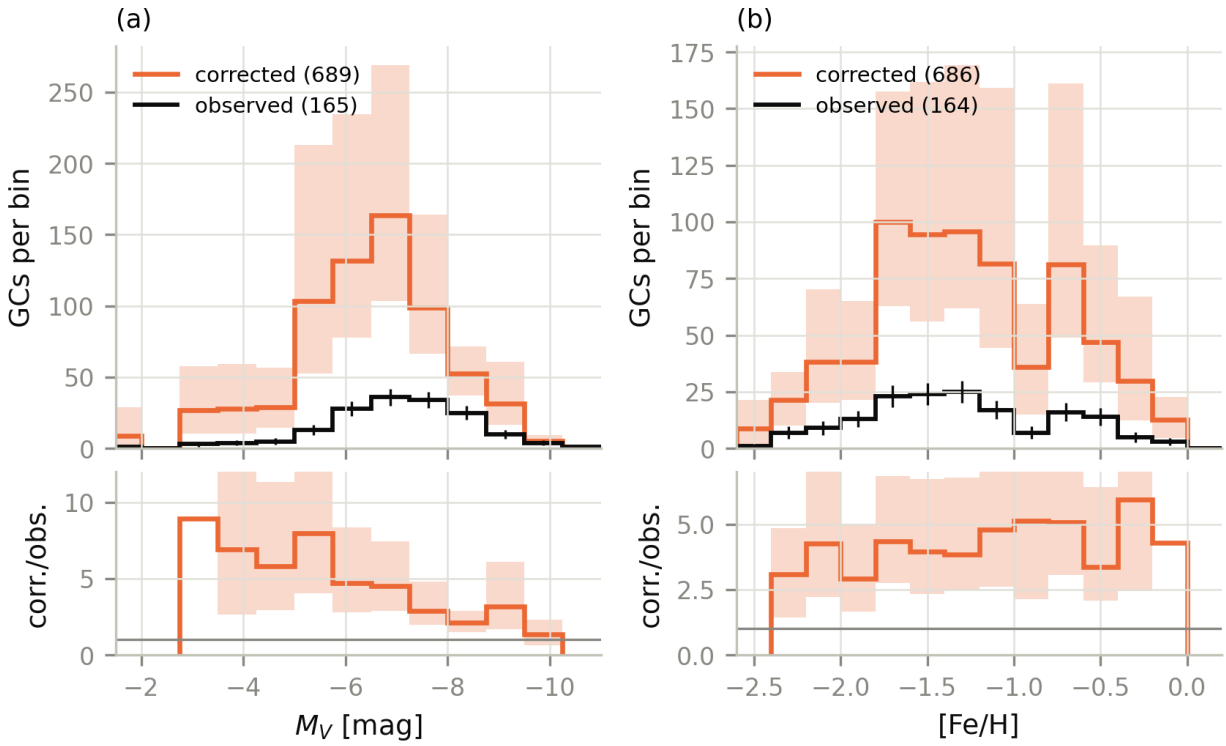


Figure 8: As Fig. 7, with $A_V = 3.1\,E(B-V)$ at the present position added to V.